3970. Shortest Path With At Most K Consecutive Identical Characters

Solved

Medium Topics Hint

You are given an integer `n` representing the number of nodes in a **directed weighted** graph, numbered from 0 to `n - 1`. This is represented by a 2D integer array `edges`, where `edges[i] = [u_i, v_i, w_i]` represents a directed edge from node u_i to node v_i with weight w_i.

You are also given a string `labels` of length `n`, where `labels[i]` is the character assigned to node `i`, and an integer `k`.

Return the **minimum total** edge weight of a path from node 0 to node `n - 1` such that the concatenation of the labels of the nodes along the path contains **at most** `k` **consecutive identical** characters. If no valid path exists, return -1.

Example 1:

> **Input:** n = 3, edges = [[0,1,1],[1,2,1],[0,2,3]], labels = "aab", k = 1
>
> **Output:** 3
>
> **Explanation:**
>
> The optimal valid path from node 0 to node 2 is as follows:
>
> - Use `edges[2] = [0, 2, 3]` to reach node 2 with a weight $w_i = 3$.
>
> The corresponding concatenation of labels is `"ab"`, which satisfies at most `k = 1` consecutive identical characters. Thus, the answer is 3.

Example 2:

> **Input:** n = 3, edges = [[0,1,1],[1,2,1],[0,2,3]], labels = "aab", k = 2
>
> **Output:** 2
>
> **Explanation:**
>
> The optimal valid path from node 0 to node 2 is as follows:
>
> - Use `edges[0] = [0, 1, 1]` to reach node 1 with weight $w_i = 1$.
> - Use `edges[1] = [1, 2, 1]` to reach node 2 with weight $w_i = 1$.
>
> The corresponding concatenation of labels is `"aab"`, which satisfies at most `k = 2` consecutive identical characters. Thus, the answer is 2.

Example 3:

> **Input:** n = 3, edges = [[0,1,1],[1,2,1]], labels = "aaa", k = 2
>
> **Output:** -1
>
> **Explanation:**
>
> There is no valid path from node 0 to node 2 that satisfies at most `k = 2` consecutive identical characters. Thus, the answer is -1.

Constraints:

- $1 <= n == labels.length <= 5 * 10^4$
- $0 <= edges.length <= 5 * 10^4$
- $edges[i] == [u_i, v_i, w_i]$
- $0 <= u_i, v_i <= n - 1$
- $u_i != v_i$
- $1 <= w_i <= 10^4$
- `labels` consists of lowercase English letters
- $1 <= k <= 50$

Seen this question in a real interview before? 1/6

Yes No

Accepted 16,905 /37.4K | Acceptance Rate 45.3 %

Topics

Hint 1

Hint 2

Hint 3

Hint 4

Discussion (42)